Lecture 1: Consumer Theory

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Utility Maximization

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1. Elements of the Consumer's Problem

Before any optimization, we must answer three questions:

1. **What can I consume?** $\rightarrow$ Defines the feasible universe of choices.
2. **What do I prefer?** $\rightarrow$ Ranks those choices.
3. **What is my objective?** $\rightarrow$ The consumer maximizes well-being subject to constraints.

1.1. The Consumption Set

The **consumption set** is

$$X = \mathbb{R}^n_+$$

It is the collection of all consumption options an individual can potentially value. An element $x \in X$ is a **consumption bundle** (basket of goods):

$$x = (x_1, x_2, \ldots, x_n)^\top \in \mathbb{R}^n_+, \qquad x_i \in \mathbb{R}_+.$$

Examples of consumption sets:

- *Standard goods list:* $x = (x_{\text{shirts}}, x_{\text{apples}}, \ldots)$.
- *Intertemporal choice:* $x = (x_1, x_2, \ldots, x_T)$, where x_t is consumption in period t.
- *Choice under uncertainty:* $x = (x_{\text{rain}}, x_{\text{sun}})$, where x_ω denotes consumption if state ω is realized.

2. Preferences

Preferences describe how the consumer ranks consumption bundles. Formally they are a **binary relation** on X.

The symbol $\succsim$ denotes the **weak preference relation**:

$$x \succsim x' \iff \text{“}x \text{ is } \textbf{at least as good as } x'\text{”}.$$

2.1. Core Axioms

For all $x, x' \in X$: $x \succsim x'$ **or** $x' \succsim x$ (or both).

Interpretation: The consumer can always compare any two bundles — decisions are always possible.

For all $x, x', x'' \in X$: if $x \succsim x'$ and $x' \succsim x''$, then $x \succsim x''$.

Interpretation: Decisions are consistent — no cycles.

A binary relation satisfying A1 and A2 is a **preference relation**.

2.2. Derived Relations

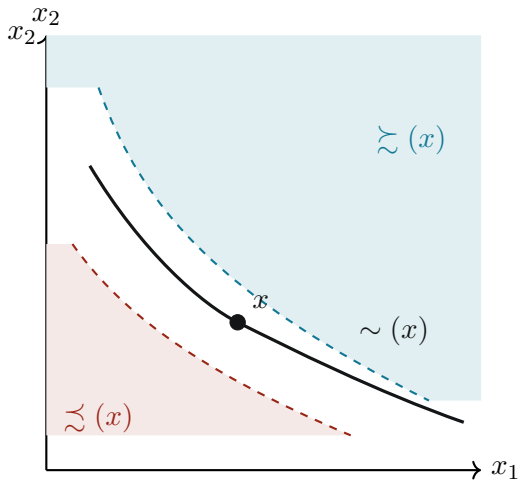
- **Indifference** ($\sim$): $x \sim x'$ if $x \succsim x'$ **and** $x' \succsim x$.
- **Strict preference** ($\succ$): $x \succ x'$ if $x \succsim x'$ **and** $x' \not\succsim x$.

Are $\sim$ and $\succ$ preference relations? They are *transitive*, but **not complete**. For instance, $\succ$ is not defined between x and x' when $x \sim x'$ — we cannot say $x \succ x'$ or $x' \succ x$.

2.3. Contour Sets

For any $x \in X$:

- $\sim(x) \equiv \{x' \in X \mid x \sim x'\}$ — **indifference set**.
- $\succsim(x) \equiv \{x' \in X \mid x' \succsim x\}$ — **upper contour set**.
- $\precsim(x) \equiv \{x' \in X \mid x \succsim x'\}$ — **lower contour set**.



3. Additional Axioms

For all $x \in X$, the sets $\succsim(x)$ and $\precsim(x)$ are **closed**.

Interpretation: Small changes in consumption produce small changes in preference rankings. No "jumps" in preferences.

For all $x, x' \in X$: if $x \geq x'$ (componentwise), then $x \succsim x'$.

Interpretation: "More is better." This eliminates thick indifference curves.

If $x \neq x'$ and $x \succsim x'$, then for all $t \in (0,1)$:

$$tx + (1-t)x' \succsim x.$$

Interpretation: The consumer prefers **mixtures** — diversification of consumption is valued.

Under strict A4, indifference curves slope downward. Under strict A5, they are strictly bowed inward (strictly convex upper contour sets).

4. The Utility Function

The utility function is a *mathematical tool* that assigns a number to each bundle so that the numerical ranking mirrors the preference ranking.

$u : \mathbb{R}^n_+ \to \mathbb{R}$ is a **utility function representing** $\succsim$ if:

$$u(x) \geq u(x') \iff x \succsim x'.$$

Existence: Under A1–A4, a utility function always exists. It is **not unique** — the numbers are **ordinal** (only their order matters, not their magnitude).

4.1. Invariance to Monotone Transformations

If $u(x)$ represents $\succsim$, then $v(x) = f(u(x))$ also represents $\succsim$ for *any* strictly increasing $f : \mathbb{R} \to \mathbb{R}$.

Example

All three functions below represent the same preferences as u:

$$\tilde{u}(x) = u(x) + 1, \qquad \hat{u}(x) = [u(x)]^3, \qquad \check{u}(x) = e^{u(x)}.$$

Good: we are free to choose the most convenient transformation.
Caveat: Numbers have no cardinal meaning — "utility of 10" vs. "utility of 5" says nothing about "twice as happy."

4.2. Properties Linking u and $\succsim$

Let $u : \mathbb{R}^n_+ \to \mathbb{R}$ represent $\succsim$. Then:

1. $u(x)$ is **strictly increasing** $\iff$ $\succsim$ is **strictly monotone**.
2. $u(x)$ is **quasiconcave** $\iff$ $\succsim$ is **convex**.
3. $u(x)$ is **strictly quasiconcave** $\iff$ $\succsim$ is **strictly convex**.

5. Quasiconcave Functions

$f : X \to \mathbb{R}$ is **quasiconcave** if for all $\alpha \in [0,1]$ and all $x^0, x^1, x^2 \in X$ with $f(x^1) \geq f(x^0)$ and $f(x^2) \geq f(x^0)$:

$$f\big(\alpha x^1 + (1-\alpha)x^2\big) \geq f(x^0).$$

$f : X \to \mathbb{R}$ is **strictly quasiconcave** if for all $\alpha \in (0,1)$, $x^1 \neq x^2$, with $f(x^1) \geq f(x^0)$ and $f(x^2) \geq f(x^0)$:

$$f\big(\alpha x^1 + (1-\alpha)x^2\big) > f(x^0).$$

The upper level sets of a quasiconcave function are convex — this is exactly why quasiconcavity of u corresponds to convexity of $\succsim$.

6. Examples of Utility Functions

6.1. Cobb–Douglas

$$u(x_1, x_2) = x_1^{\alpha} x_2^{\beta}, \quad \alpha, \beta > 0.$$

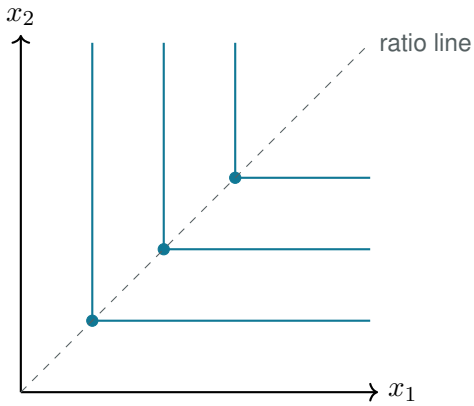
General case (n goods): $u(x) = \prod_{i=1}^{n} x_i^{\alpha_i}$.

Indifference curves are smooth, strictly convex, and asymptotic to the axes.

6.2. Perfect Complements

$$u(x) = \min\{k_1 x_1,\ k_2 x_2,\ \ldots,\ k_n x_n\}, \quad k_i > 0.$$

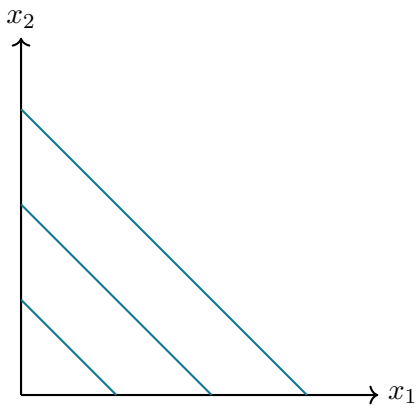
Indifference curves are **L-shaped**. The consumer consumes goods in fixed proportions $k_1 x_1 = k_2 x_2 = \cdots$; extra units of one good (beyond the required ratio) yield no additional utility.



6.3. Perfect Substitutes

$$u(x) = k_1x_1 + k_2x_2 + \cdots + k_nx_n.$$

Indifference curves are **straight lines** with slope $-k_1/k_2$. The consumer is willing to substitute goods at a constant rate regardless of how many of each she already has.



7. Marginal Utility and the Marginal Rate of Substitution

7.1. Marginal Utility

$$MU_i(x) \equiv \frac{\partial u(x)}{\partial x_i}.$$

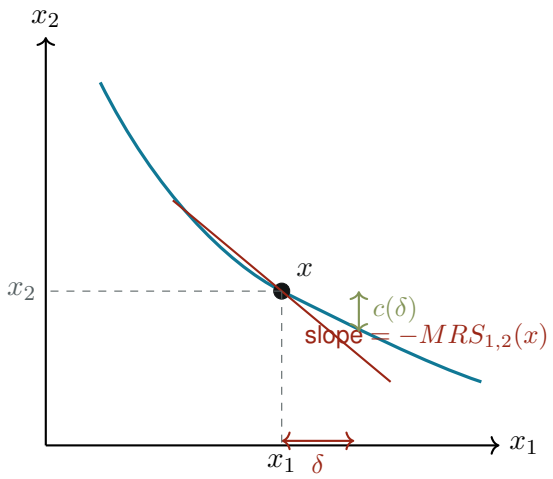
This is the rate of change of utility from consuming one more unit of good i, holding all other quantities fixed.

7.2. Marginal Rate of Substitution (MRS)

$$MRS_{i,j}(x) \equiv \frac{\partial u(x)/\partial x_i}{\partial u(x)/\partial x_j}.$$

It answers: *how much of good j must I receive to compensate for losing one unit of good i, while remaining on the same indifference curve?*

Geometrically (for $n = 2$), $MRS_{1,2}$ is the **absolute slope** of the indifference curve at x. A small reduction δ in x_1 requires an increase $c(\delta) = MRS_{1,2}(x) \cdot \delta$ in x_2 to maintain the same utility level.

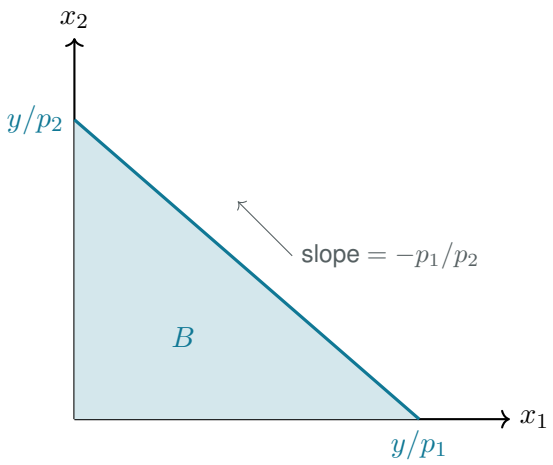


8. Feasible Options: The Budget Set

Given **prices** $p = (p_1, \ldots, p_n) \in \mathbb{R}^n_+$, $p > 0$, and **income** $y \in \mathbb{R}_+$, $y > 0$, the consumer's **budget set** is:

$$B = \{x \in X \mid p \cdot x \leq y\}.$$

Agents are **price-takers**: they take p as given. The boundary of B is the **budget line**: $p_1x_1 + p_2x_2 = y$ (for $n = 2$), with intercepts y/p_1 and y/p_2 and slope $-p_1/p_2$.



9. The Consumer's Problem (UMP)

Behavioral assumption: The consumer selects the *best available* bundle — the one that is $\succsim$-maximal in B.

$$\max_{x \in \mathbb{R}^n_+} \; u(x) \quad \text{subject to} \quad p \cdot x \leq y$$

9.1. Existence of a Solution

For a solution to the UMP to exist we need:

1. $p \gg 0$: all prices strictly positive, so B is bounded. (If $p_i = 0$, demand for good i could be $+\infty$.)
2. Continuity of u (ensured by A3).

A continuous function attains its maximum on a compact (closed and bounded) set. Under $p \gg 0$ and continuity of u, the budget set B is compact, so a solution to the UMP exists.

9.2. Uniqueness of the Solution

Under A1–A3, *strict* monotonicity, and *strict* convexity (A5), the solution to the UMP is **unique**.

Proof. By contradiction. Suppose $x_1 \neq x_2$ are both solutions. Then:

$$p \cdot x_1 \leq y, \quad p \cdot x_2 \leq y, \quad u(x_1) = u(x_2).$$

Define $x_3 = \frac{1}{2}x_1 + \frac{1}{2}x_2$. This is feasible:

$$p \cdot x_3 = \tfrac{1}{2}(p \cdot x_1 + p \cdot x_2) \leq y.$$

By **strict quasiconcavity** of u:

$$u(x_3) > u(x_1) = u(x_2).$$

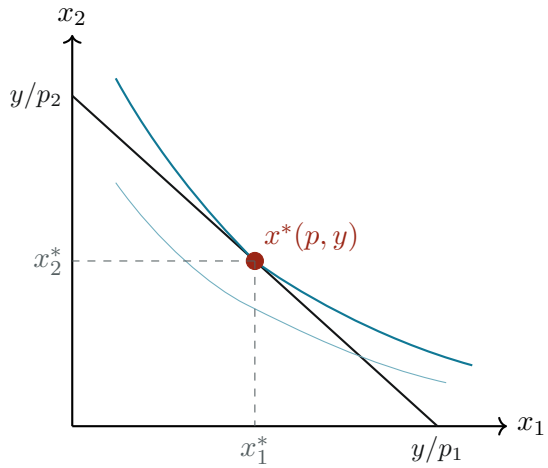
This contradicts x_1 and x_2 being optimal. Intuitively, we can always construct a strictly better convex combination. ■

10. The Marshallian Demand Function

The solution to the UMP defines the **Marshallian (ordinary/uncompensated) demand**:

$$x^*(p, y) = \arg\max_{x \in \mathbb{R}^n_+} u(x) \quad \text{s.t.} \quad p \cdot x \leq y.$$

This is a function (under strict convexity) $x^d : \mathbb{R}^n_+ \times \mathbb{R}_+ \to \mathbb{R}^n_+$ mapping prices and income to demanded quantities.



11. First-Order Conditions: Lagrangian Approach

Assuming u is differentiable, form the Lagrangian:

$$\mathcal{L}(x,\lambda) = u(x) - \lambda\,(p \cdot x - y).$$

Kuhn–Tucker conditions:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial u(x^*)}{\partial x_i} - \lambda^* p_i = 0, \qquad \forall\, i,$$

$$\lambda^*(p \cdot x^* - y) \leq 0.$$

By **strict monotonicity** the budget constraint binds, $p \cdot x^* = y$, so:

$$\boxed{\frac{\partial u(x^*)}{\partial x_i} = \lambda^* p_i} \qquad \text{(MU proportional to price)}$$

Taking the ratio for goods i and j:

$$\boxed{MRS_{i,j}(x^*) = \frac{p_i}{p_j}} \qquad \text{(slope of IC = price ratio at optimum)}$$

For $n = 2$: the indifference curve is tangent to the budget line at the optimum.

11.1. CES Example: Marshallian Demand

Consider the **Constant Elasticity of Substitution (CES)** utility:

$$u(x_1, x_2) = \left(x_1^\rho + x_2^\rho\right)^{1/\rho}, \qquad 0 \neq \rho < 1.$$

The Lagrangian is:

$$\mathcal{L}(x_1, x_2, \lambda) = (x_1^\rho + x_2^\rho)^{1/\rho} - \lambda(p_1 x_1 + p_2 x_2 - y).$$

First-order conditions:

$$[x_1]: \quad \tfrac{1}{\rho}\,(x_1^\rho + x_2^\rho)^{1/\rho - 1}\,\rho\, x_1^{\rho-1} - \lambda p_1 = 0 \tag{1}$$

$$[x_2]: \quad \tfrac{1}{\rho}\,(x_1^\rho + x_2^\rho)^{1/\rho - 1}\,\rho\, x_2^{\rho-1} - \lambda p_2 = 0 \tag{2}$$

$$[\lambda]: \quad p_1 x_1 + p_2 x_2 = y \tag{3}$$

Dividing (1) by (2):

$$\left(\frac{x_1}{x_2}\right)^{\rho-1} = \frac{p_1}{p_2} \implies x_1 = \left(\frac{p_1}{p_2}\right)^{\frac{1}{\rho-1}} x_2.$$

Substituting into (3) and letting $r \equiv \dfrac{\rho}{\rho - 1}$, we obtain the **Marshallian demand**:

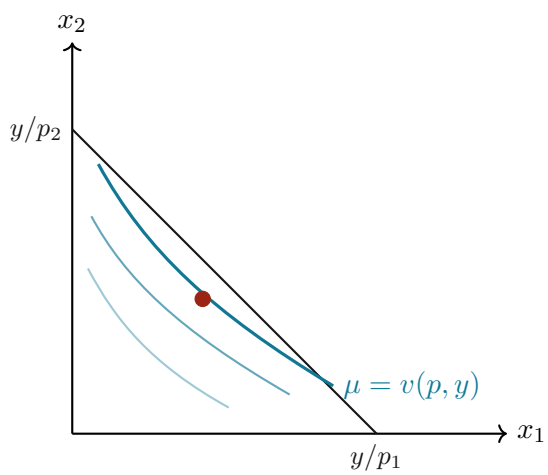
$$\boxed{x_i^*(p, y) = \frac{y\, p_i^{r-1}}{p_1^r + p_2^r}, \qquad i = 1, 2.}$$

12. Indirect Utility Function

$$v(p, y) \;=\; \max_{x \in X}\; u(x) \quad \text{s.t.} \quad p \cdot x \leq y.$$

Equivalently: $v(p, y) = u(x^*(p, y))$.

Idea: Fix prices p and income y; ask what is the *maximum achievable utility*?



12.1. Properties of the Indirect Utility Function

Under A1–A5, $v(p, y)$ is:

1. **Continuous** in (p, y).
2. **Homogeneous of degree 0** in (p, y): $v(tp, ty) = v(p, y)$ for all $t > 0$.
3. **Strictly increasing** in y.
4. **Decreasing** (non-increasing) in p.
5. **Quasiconcave** in (p, y).
6. Satisfies **Roy's Identity**.

12.2. Roy's Identity

If $v(p, y)$ is differentiable at (p^0, y^0) and $\partial v(p^0, y^0)/\partial y \neq 0$, then:

$$x_i(p^0, y^0) = -\frac{\partial v(p^0, y^0)/\partial p_i}{\partial v(p^0, y^0)/\partial y}, \qquad i = 1, \ldots, n.$$

Proof. Apply the **Envelope Theorem** to $v(p, y) = \max_x \mathcal{L}$ where $\mathcal{L}(x, \lambda) = u(x) - \lambda(p \cdot x - y)$:

$$\frac{\partial v}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial p_i}\bigg|_{x^*, \lambda^*} = -\lambda^* x_i^*, \qquad \frac{\partial v}{\partial y} = \frac{\partial \mathcal{L}}{\partial y}\bigg|_{x^*, \lambda^*} = \lambda^*.$$

Therefore:

$$-\frac{\partial v/\partial p_i}{\partial v/\partial y} = \frac{\lambda^* x_i^*}{\lambda^*} = x_i^*. \qquad \square$$

■

12.3. CES Example: Indirect Utility

For the CES utility of Section 11.1, substituting the Marshallian demands yields:

$$v(p, y) = y\left(p_1^r + p_2^r\right)^{-1/r}.$$

Verify all six properties of $v(p, y)$ for the CES example above.

13. Expenditure Function and Hicksian Demand

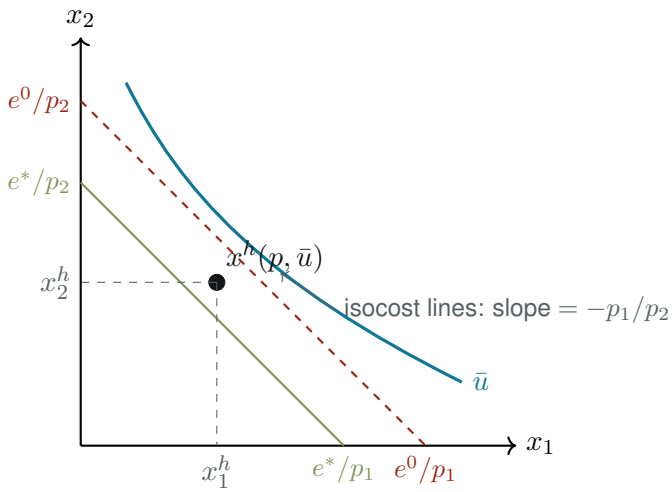
The **dual** of the UMP is expenditure minimization.

Given prices $p \gg 0$ and a utility target $\bar{u}$, find the cheapest bundle achieving $\bar{u}$:

$$e(p, \bar{u}) \equiv \min_{x \in X} p \cdot x \quad \text{s.t.} \quad u(x) \geq \bar{u}.$$

The solution $x^h(p, \bar{u})$ is the **Hicksian (compensated) demand**. The expenditure function satisfies $e(p, \bar{u}) = p \cdot x^h(p, \bar{u})$.

Interpretation: Suppose prices rise. Normally the consumer's real income falls. Here we *compensate* her with enough income to keep utility at $\bar{u}$.



13.1. Properties of the Expenditure Function

1. **Homogeneous of degree 1** in p: $e(tp, \bar{u}) = t\, e(p, \bar{u})$ for $t > 0$.
2. **Increasing** in p.
3. **Concave** in p.
4. **Shephard's Lemma**: if e is differentiable in p,

$$\frac{\partial e(p, \bar{u})}{\partial p_i} = x_i^h(p, \bar{u}), \qquad i = 1, \ldots, n.$$

Proof (Shephard's Lemma). The EMP Lagrangian is $\mathcal{L} = p \cdot x - \gamma(u(x) - \bar{u})$. By the Envelope Theorem:

$$\frac{\partial e(p, \bar{u})}{\partial p_i} = \left.\frac{\partial \mathcal{L}}{\partial p_i}\right|_{x^*, \gamma^*} = x_i^* \equiv x_i^h(p, \bar{u}). \qquad \square$$

■

13.2. CES Example: Hicksian Demand and Expenditure Function

For $u(x_1, x_2) = (x_1^\rho + x_2^\rho)^{1/\rho}$, the EMP first-order conditions give:

$$x_i^h(p, \bar{u}) = \bar{u} \left(\frac{p_i^r}{p_1^r + p_2^r} \right)^{1/\rho}, \qquad i = 1, 2,$$

and the expenditure function is:

$$e(p, \bar{u}) = \bar{u} \left(p_1^r + p_2^r \right)^{1/r}.$$

14. Duality: Relating v and e

Let $v(p, y)$ and $e(p, \bar{u})$ be the indirect utility and expenditure functions for a consumer with continuous, strictly increasing utility. Then for all $p \gg 0$, $y \geq 0$:

1. $e(p,\, v(p, y)) = y$.
2. $v(p,\, e(p, \bar{u})) = \bar{u}$.

Interpretation: The two functions are inverses of each other with respect to the income/utility argument. Starting from utility maximization and switching to expenditure minimization (and back) returns you to the starting point.

1. $x_i(p, y) = x_i^h(p,\, v(p, y))$.
2. $x_i^h(p, \bar{u}) = x_i(p,\, e(p, \bar{u}))$.

Proof. Let $x^0 = x(p^0, y^0)$, $u^0 = u(x^0)$. By definition $v(p^0, y^0) = u^0$ and $p^0 \cdot x^0 = y^0$, so $e(p^0, u^0) = y^0$. Now x^0 solves the EMP at (p^0, u^0), so $x^0 = x^h(p^0, u^0)$. The second statement follows symmetrically. □ ■

15. Properties of Marshallian Demand

15.1. Homogeneity and Budget Balance

Under A1–A5, the Marshallian demand $x_i(p, y)$ satisfies:

1. **Homogeneous of degree 0** in (p, y): $\quad x_i(tp, ty) = x_i(p, y), \quad \forall\, t > 0.$
2. **Walras's Law**: $\quad p \cdot x(p, y) = y.$

Interpretation: Only **relative prices** p_i/p_j and **real income** y/p_j affect behavior. Setting $t = 1/p_n$:

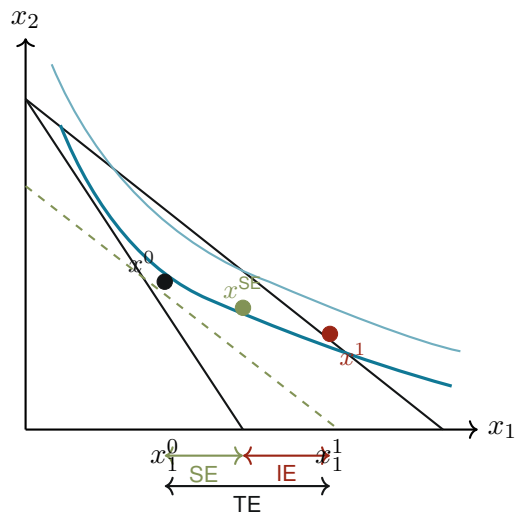
$$x(p, y) = x\left(\frac{p_1}{p_n}, \ldots, \frac{p_{n-1}}{p_n}, 1, \frac{y}{p_n} \right).$$

The demand for each good depends on $n - 1$ relative prices and real income only.

15.2. Income Effect and Substitution Effect

When price p_i falls, two things happen simultaneously:

- **Substitution Effect (SE):** Good i becomes relatively cheaper compared to other goods. Measured by the Hicksian demand: fix $\bar{u}$ and track how demand changes with p_i alone. Depends *only* on relative prices.
- **Income Effect (IE):** The price fall raises real purchasing power. It is the change in income needed to compensate the consumer for the utility change.



16. The Slutsky Equation

$$\underbrace{\frac{\partial x_i(p,y)}{\partial p_j}}_{\text{Total Effect (TE)}} = \underbrace{\frac{\partial x_i^h(p,u^*)}{\partial p_j}}_{\text{Substitution Effect (SE)}} - \underbrace{x_j(p,y)\,\frac{\partial x_i(p,y)}{\partial y}}_{\text{Income Effect (IE)}},$$

where $u^* = v(p,y)$ is the consumer's maximized utility.

Proof. Recall $x_i^h(p,u^*) = x_i(p, e(p,u^*))$. Differentiate with respect to p_j:

$$\frac{\partial x_i^h}{\partial p_j} = \frac{\partial x_i(p,e(p,u^*))}{\partial p_j} + \frac{\partial x_i(p,e(p,u^*))}{\partial y} \cdot \frac{\partial e(p,u^*)}{\partial p_j}.$$

By Shephard's Lemma, $\partial e/\partial p_j = x_j^h = x_j(p,y)$, and $e(p,u^*) = y$. Rearranging gives the Slutsky equation. □ ■

17. Law of Demand

A reduction in the price of a **normal** good causes an increase in quantity demanded. If a price reduction causes quantity demanded to *fall*, the good is **inferior**.

Proof. For a normal good $\partial x_i/\partial y > 0$. From the Slutsky equation with $i = j$:

$$\frac{\partial x_i}{\partial p_i} = \underbrace{\frac{\partial x_i^h}{\partial p_i}}_{\leq 0} - \underbrace{x_i(p,y)}_{>0} \underbrace{\frac{\partial x_i}{\partial y}}_{>0} \leq 0. \qquad \square$$

■

- **Normal good:** $\dfrac{\partial x_i(p,y)}{\partial y} > 0.$
- **Inferior good:** $\dfrac{\partial x_i(p,y)}{\partial y} < 0.$

18. The Slutsky Matrix

18.1. Symmetry of Hicksian Cross-Derivatives

$$\frac{\partial x_i^h(p,\bar{u})}{\partial p_j} = \frac{\partial x_j^h(p,\bar{u})}{\partial p_i}.$$

Proof. By Shephard's Lemma $x_i^h = \partial e/\partial p_i$, so $\partial x_i^h/\partial p_j = \partial^2 e/(\partial p_i\, \partial p_j) = \partial^2 e/(\partial p_j\, \partial p_i) = \partial x_j^h/\partial p_i$. $\square$ ■

18.2. The Substitution Matrix

Collect all Hicksian price derivatives into the **substitution matrix**:

$$\sigma(p,\bar{u}) = \left(\frac{\partial x_i^h(p,\bar{u})}{\partial p_j}\right)_{i,j} = \nabla_p^2\, e(p,\bar{u}).$$

Since e is concave in p, $\sigma(p,\bar{u})$ is **negative semidefinite** (NSD):

$$\sigma = \nabla_p^2\, e(p,\bar{u}) \text{ is NSD} \implies \frac{\partial x_i^h}{\partial p_i} \leq 0 \quad \forall\, i.$$

The **Slutsky matrix** $S(p,y)$ with (i,j)-element $\partial x_i/\partial p_j + x_j \cdot \partial x_i/\partial y$ is also **negative semidefinite**.

19. Elasticities

$$\eta_i \equiv \frac{\partial x_i(p,y)}{\partial y} \cdot \frac{y}{x_i(p,y)}.$$

Percentage change in demand for good i per 1% increase in income.

$$\varepsilon_{ij} \equiv \frac{\partial x_i(p,y)}{\partial p_j} \cdot \frac{p_j}{x_i(p,y)}.$$

Percentage change in demand for good i per 1% increase in price of good j. When $i = j$: **own-price elasticity**. When $i \neq j$: **cross-price elasticity**.

Reference: Jehle & Reny, Advanced Microeconomic Theory*, Chapters 1–2.*